

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B. Tech CSE & SPL

Semester: VI

Name of the Course: Security and Auditing

Course Code: TCS 695

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1 (20 Marks)		
(a)	Examine the impact of ransomware on global businesses. How can organizations implement proactive strategies to avoid becoming victims of ransomware attacks?	CO2
(b)	What role do botnets play in distributed denial-of-service (DDoS) attacks? Explain how bot networks are created, controlled, and monetized by attackers.	CO1
(c)	Describe the working procedure of a phishing email. What techniques are commonly used to convince recipients to click malicious links or submit sensitive information? Use figures in your answer.	CO1
Q2 (20 Marks)		
(a)	What do you understand about Spam over Instant Messaging (SPIM)? Give one example. How can organizations create and enforce an instant messaging security policy that minimizes the risks posed by SPIM?	CO3
(b)	How do Signature-based detection and Anomaly-based detection methods in IDS/IPS systems work? Discuss the pros and cons of each, and explain the challenges associated with maintaining high detection accuracy in both systems.	CO3
(c)	In a typical cybersecurity framework, what are the differences between operational, tactical, and strategic roles? Give examples for each.	CO2
Q3 (20 Marks)		
(a)	How do modern firewalls differentiate between legitimate and malicious traffic? Discuss the key differences between packet-filtering firewalls, proxy firewalls and Firewall as a Service (FWaaS).	CO4
(b)	What is General Data Protection Regulation (GDPR). What is the impact of GDPR on Indian IT companies handling European citizens' data?	CO2
(c)	Describe the different types of Virtual Private Networks (VPNs): site-to-site and point-to-point. In what contexts is each type most appropriate, and how do they differ in terms of configuration, security, and performance?	CO5

Q4	(20 Marks)	
(a)	How does the design of Wi-Fi Protected Access (WPA2) contribute to wireless network security? Explain the vulnerabilities of WPA2 and how WPA3 addresses these weaknesses.	CO5
(b)	What are the common challenges in database auditing and monitoring? Provide the details of the following tools used for the database auditing and monitoring. IBM Guardium, Oracle Audit Vault and SQL Server Audit.	CO6
(c)	What is the use of audit trails and logs? What are the different types of logs? Demonstrate the working procedure of audit trails and logs.	CO4
Q5	(20 Marks)	
(a)	Why it is important to perform database vulnerability assessment and penetration testing. What are the different steps of database vulnerability assessment and penetration testing?	CO3
(b)	Compare and contrast full backups, incremental backups, and differential backups. What are the advantages and disadvantages of each, and in what situations would each type be most appropriate?	CO6
(c)	An e-commerce company performs database backups every 2 hours. The system crashes at 3:30 PM due to a power failure. Find out the value of Recovery Point Objective (RPO), i.e. how much data might be lost? Further find out the value of Recovery Time Objective (RTO), if the company's IT team takes 1.5 hours to restore the system, when will the company be operational again?	CO5